$$
\begin{aligned}
x_1 &= L_1 \sin\theta_1 \\
y_1 &= L_1 \cos\theta_1
\end{aligned}
$$

$$
\begin{aligned}
x_2 &= L_1 \sin\theta_1 + L_2 \sin\theta_2 \\
y_2 &= L_1 \cos\theta_1 + L_2 \cos\theta_2
\end{aligned}
$$

$$
\begin{aligned}
\dot{x}_1 &= L_1 \dot{\theta}_1 \cos\theta_1 \\
\dot{y}_1 &= -L_1 \dot{\theta}_1 \sin\theta_1
\end{aligned}
$$

$$
\begin{aligned}
\dot{x}_2 &= L_1 \dot{\theta}_1 \cos\theta_1 + L_2 \dot{\theta}_2 \cos\theta_2 \\
\dot{y}_2 &= -L_1 \dot{\theta}_1 \sin\theta_1 - L_2 \dot{\theta}_2 \sin\theta_2
\end{aligned}
$$

$$
\begin{aligned}
v_1^2 &= \dot{x}_1^2 + \dot{y}_1^2 \\
&= L_1^2 \dot{\theta}_1^2 \cos^2\theta_1 + L_1^2 \dot{\theta}_1^2 \sin^2\theta_1 \\
&= L_1^2 \dot{\theta}_1^2
\end{aligned}
$$

$$
\begin{aligned}
v_2^2 &= \dot{x}_2^2 + \dot{y}_2^2 \\
&= (L_1 \dot{\theta}_1 \cos\theta_1 + L_2 \dot{\theta}_2 \cos\theta_2)^2 + (L_1 \dot{\theta}_1 \sin\theta_1 + L_2 \dot{\theta}_2 \sin\theta_2)^2 \\
&= L_1^2 \dot{\theta}_1^2 \cos^2\theta_1 + 2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos\theta_1 \cos\theta_2 + L_2^2 \dot{\theta}_2^2 \cos^2\theta_2 \\
&\quad + L_1^2 \dot{\theta}_1^2 \sin^2\theta_1 + 2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \sin\theta_1 \sin\theta_2 + L_2^2 \dot{\theta}_2^2 \sin^2\theta_2 \\
&= L_1^2 \dot{\theta}_1^2 + L_2^2 \dot{\theta}_2^2 + 2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 (\cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2) \\
&= L_1^2 \dot{\theta}_1^2 + L_2^2 \dot{\theta}_2^2 + 2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)
\end{aligned}
$$

$$
\begin{aligned}
T &= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \\
&= \frac{1}{2} m_1 L_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 \left[L_1^2 \dot{\theta}_1^2 + L_2^2 \dot{\theta}_2^2 + 2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right] \\
&= \frac{1}{2} (m_1 + m_2) L_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 L_2^2 \dot{\theta}_2^2 + m_2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)
\end{aligned}
$$

$$
\begin{aligned}
V &= -m_1 g y_1 - m_2 g y_2 \\
&= -m_1 g L_1 \cos\theta_1 - m_2 g (L_1 \cos\theta_1 + L_2 \cos\theta_2) \\
&= -(m_1 + m_2) g L_1 \cos\theta_1 - m_2 g L_2 \cos\theta_2
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L} &= T - V \\
&= \frac{1}{2} (m_1 + m_2) L_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 L_2^2 \dot{\theta}_2^2 + m_2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\
&\quad + (m_1 + m_2) g L_1 \cos\theta_1 + m_2 g L_2 \cos\theta_2
\end{aligned}
$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$

Let $\delta = \theta_1 - \theta_2$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} = (m_1 + m_2)L_1^2\dot{\theta}_1 + m_2 L_1 L_2 \dot{\theta}_2 \cos\delta$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1}\right) = (m_1 + m_2)L_1^2\ddot{\theta}_1 + m_2 L_1 L_2 \ddot{\theta}_2 \cos\delta - m_2 L_1 L_2 \dot{\theta}_2(\dot{\theta}_1 - \dot{\theta}_2)\sin\delta$$

$$\frac{\partial \mathcal{L}}{\partial \theta_1} = -m_2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \sin\delta - (m_1 + m_2) g L_1 \sin\theta_1$$

$$\begin{aligned}(m_1 + m_2)L_1^2\ddot{\theta}_1 + m_2 L_1 L_2 \ddot{\theta}_2 \cos\delta - m_2 L_1 L_2 \dot{\theta}_2(\dot{\theta}_1 - \dot{\theta}_2)\sin\delta \\ + m_2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \sin\delta + (m_1 + m_2) g L_1 \sin\theta_1 = 0\end{aligned}$$

$$(m_1 + m_2)L_1\ddot{\theta}_1 + m_2 L_2 \ddot{\theta}_2 \cos\delta + m_2 L_2 \dot{\theta}_2^2 \sin\delta + (m_1 + m_2) g \sin\theta_1 = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} = m_2 L_2^2 \dot{\theta}_2 + m_2 L_1 L_2 \dot{\theta}_1 \cos\delta$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_2}\right) = m_2 L_2^2 \ddot{\theta}_2 + m_2 L_1 L_2 \ddot{\theta}_1 \cos\delta + m_2 L_1 L_2 \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2) \sin\delta$$

$$\frac{\partial \mathcal{L}}{\partial \theta_2} = m_2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \sin\delta - m_2 g L_2 \sin\theta_2$$

$$\begin{aligned} &m_2 L_2^2 \ddot{\theta}_2 + m_2 L_1 L_2 \ddot{\theta}_1 \cos\delta + m_2 L_1 L_2 \dot{\theta}_1 (\dot{\theta}_1 - \dot{\theta}_2) \sin\delta \\ &\quad - m_2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \sin\delta + m_2 g L_2 \sin\theta_2 = 0 \end{aligned}$$

$$m_2 L_2 \ddot{\theta}_2 + m_2 L_1 \ddot{\theta}_1 \cos\delta - m_2 L_1 \dot{\theta}_1^2 \sin\delta + m_2 g \sin\theta_2 = 0$$

$$\begin{pmatrix} (m_1+m_2)L_1 & m_2L_2\cos\delta \\ m_2L_1\cos\delta & m_2L_2 \end{pmatrix} \begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} = \begin{pmatrix} -m_2L_2\dot{\theta}_2^2\sin\delta - (m_1+m_2)g\sin\theta_1 \\ m_2L_1\dot{\theta}_1^2\sin\delta - m_2g\sin\theta_2 \end{pmatrix}$$

$$\begin{aligned} D &= (m_1+m_2)L_1 \cdot m_2L_2 - m_2^2L_1L_2\cos^2\delta \\ &= m_2L_1L_2\left[(m_1+m_2) - m_2\cos^2\delta\right] \\ &= m_2L_1L_2\left[m_1 + m_2(1-\cos^2\delta)\right] \\ &= m_2L_1L_2\left[m_1 + m_2\sin^2\delta\right] \end{aligned}$$

$$\sin^2\delta = \frac{1-\cos 2\delta}{2}$$

$$D = \frac{m_2L_1L_2}{2}\left(2m_1 + m_2 - m_2\cos 2\delta\right)$$

$$\ddot{\theta}_1 = \frac{1}{D}\det\begin{pmatrix} -m_2L_2\dot{\theta}_2^2\sin\delta - (m_1+m_2)g\sin\theta_1 & m_2L_2\cos\delta \\ m_2L_1\dot{\theta}_1^2\sin\delta - m_2g\sin\theta_2 & m_2L_2 \end{pmatrix}$$

$$= \frac{1}{D}\Big[\Big(-m_2L_2\dot{\theta}_2^2\sin\delta - (m_1+m_2)g\sin\theta_1\Big)(m_2L_2) - (m_2L_2\cos\delta)\Big(m_2L_1\dot{\theta}_1^2\sin\delta - m_2g\sin\theta_2\Big)\Big]$$

$$= \frac{1}{D}\Big[-m_2^2L_2^2\dot{\theta}_2^2\sin\delta - (m_1+m_2)m_2gL_2\sin\theta_1 - m_2^2L_1L_2\dot{\theta}_1^2\sin\delta\cos\delta + m_2^2gL_2\cos\delta\sin\theta_2\Big]$$

$$= \frac{m_2L_2}{D}\Big[-m_2L_2\dot{\theta}_2^2\sin\delta - (m_1+m_2)g\sin\theta_1 - m_2L_1\dot{\theta}_1^2\sin\delta\cos\delta + m_2g\cos\delta\sin\theta_2\Big]$$

$$= \frac{2}{L_1(2m_1+m_2-m_2\cos 2\delta)}\Big[-m_2L_2\dot{\theta}_2^2\sin\delta - (m_1+m_2)g\sin\theta_1 - m_2L_1\dot{\theta}_1^2\sin\delta\cos\delta + m_2g\cos\delta\sin\theta_2\Big]$$

$$\cos\delta\sin\theta_2 = \cos(\theta_1-\theta_2)\sin\theta_2 = \frac{1}{2}\left[\sin\theta_1 + \sin(2\theta_2-\theta_1)\right]$$

$$\sin\theta_1 - \frac{1}{2}\sin\theta_1 - \frac{1}{2}\sin(2\theta_2-\theta_1) = \frac{1}{2}\sin\theta_1 - \frac{1}{2}\sin(2\theta_2-\theta_1) = -\frac{1}{2}\left[\sin(\theta_1-2\theta_2) + \sin\theta_1\right]$$

$$\ddot{\theta}_1 = \frac{-g(2m_1+m_2)\sin\theta_1 - m_2g\sin(\theta_1-2\theta_2) - 2\sin\delta\cdot m_2(\dot{\theta}_2^2L_2 + \dot{\theta}_1^2L_1\cos\delta)}{L_1(2m_1+m_2-m_2\cos 2\delta)}$$

$$\ddot{\theta}_2 = \frac{1}{D}\det\begin{pmatrix} (m_1+m_2)L_1 & -m_2L_2\dot{\theta}_2^2\sin\delta - (m_1+m_2)g\sin\theta_1 \\ m_2L_1\cos\delta & m_2L_1\dot{\theta}_1^2\sin\delta - m_2g\sin\theta_2 \end{pmatrix}$$

$$= \frac{1}{D}\Big[(m_1+m_2)L_1\left(m_2L_1\dot{\theta}_1^2\sin\delta - m_2g\sin\theta_2\right) - (m_2L_1\cos\delta)\left(-m_2L_2\dot{\theta}_2^2\sin\delta - (m_1+m_2)g\sin\theta_1\right)\Big]$$

$$= \frac{1}{D}\Big[(m_1+m_2)m_2L_1^2\dot{\theta}_1^2\sin\delta - (m_1+m_2)m_2gL_1\sin\theta_2 + m_2^2L_1L_2\dot{\theta}_2^2\sin\delta\cos\delta + (m_1+m_2)m_2gL_1\cos\delta\sin\theta_1\Big]$$

$$= \frac{m_2L_1}{D}\Big[(m_1+m_2)L_1\dot{\theta}_1^2\sin\delta - (m_1+m_2)g\sin\theta_2 + m_2L_2\dot{\theta}_2^2\sin\delta\cos\delta + (m_1+m_2)g\cos\delta\sin\theta_1\Big]$$

$$= \frac{2}{L_2(2m_1+m_2-m_2\cos 2\delta)}\Big[(m_1+m_2)L_1\dot{\theta}_1^2\sin\delta + m_2L_2\dot{\theta}_2^2\sin\delta\cos\delta + (m_1+m_2)g(\cos\delta\sin\theta_1 - \sin\theta_2)\Big]$$

$$\cos\delta\sin\theta_1 - \sin\theta_2 = \cos(\theta_1-\theta_2)\sin\theta_1 - \sin\theta_2 = \sin\delta\cos\theta_1$$

$$\ddot{\theta}_2 = \frac{2\sin\delta\left[\dot{\theta}_1^2L_1(m_1+m_2) + g(m_1+m_2)\cos\theta_1 + \dot{\theta}_2^2L_2m_2\cos\delta\right]}{L_2(2m_1+m_2-m_2\cos 2\delta)}$$